U.S. Hospital Quality, Spending & Readmission Risk Analyzer

3.12
Star rating

11.00
Readmissions Rate

3.12
Safe Rating

7.00
Mortality Rate

90%
Patient Experience

Search Hospital Name

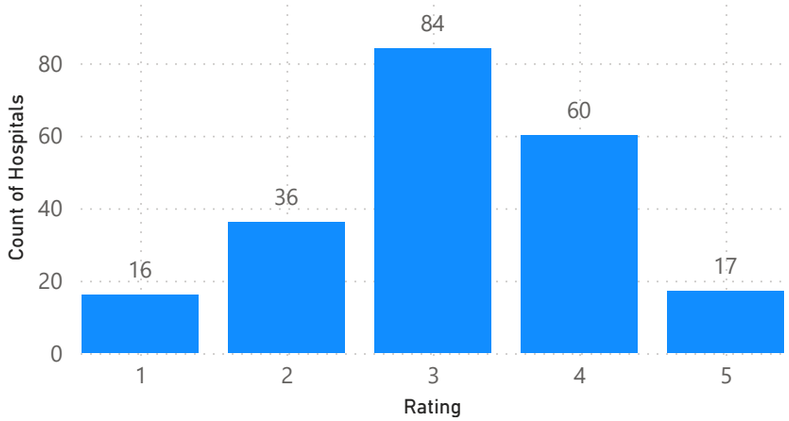
All

Hospital Name by Full Location

Hospital type

- [ ] Acute Care - Department of Defense
- [ ] Acute Care - Veterans Administration
- [ ] Acute Care Hospitals
- [ ] Childrens
- [x] Critical Access Hospitals
- [ ] Long-term
- [ ] Psychiatric
- [ ] Rural Emergency Hospital

Count of Hospitals by Rating



Dashboard Methodology & Key Findings

This analyzer explores the relationship between high-level hospital ratings and specific clinical outcomes.

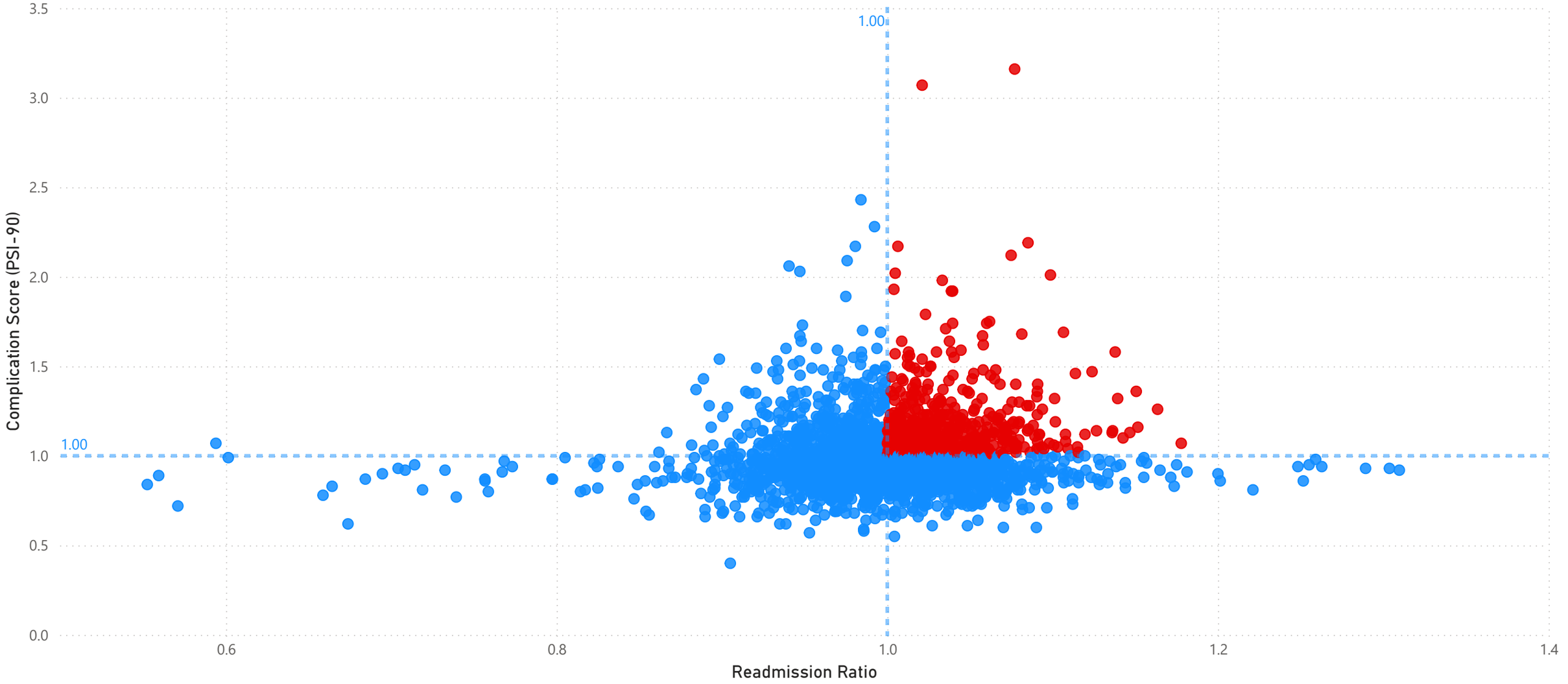
Metric Definitions:

- **Star Rating:** The CMS summary of overall quality.
- **Readmission & Mortality:** Scores centered around **1.0 (National Average)**.
  - **< 1.0:** Better than average (Lower risk).
  - **> 1.0:** Worse than average (Higher risk).
- **Patient Experience:** Percentage of positive patient survey responses.

Observed Correlations: Analysis of this dataset confirms that **CMS Star Ratings are highly aligned with actual clinical outcomes.** * **5-Star Hospitals** consistently demonstrate **lower readmission ratios** and significantly **higher patient experience scores**.

- High-rated facilities also show improved performance in **mortality rates and safety measures**, consistently staying below the national risk average.

Hospital Risk Analysis: Readmissions vs. Complications



ABBEVILLE GENERAL HOSPITAL

First Hospital

1.00

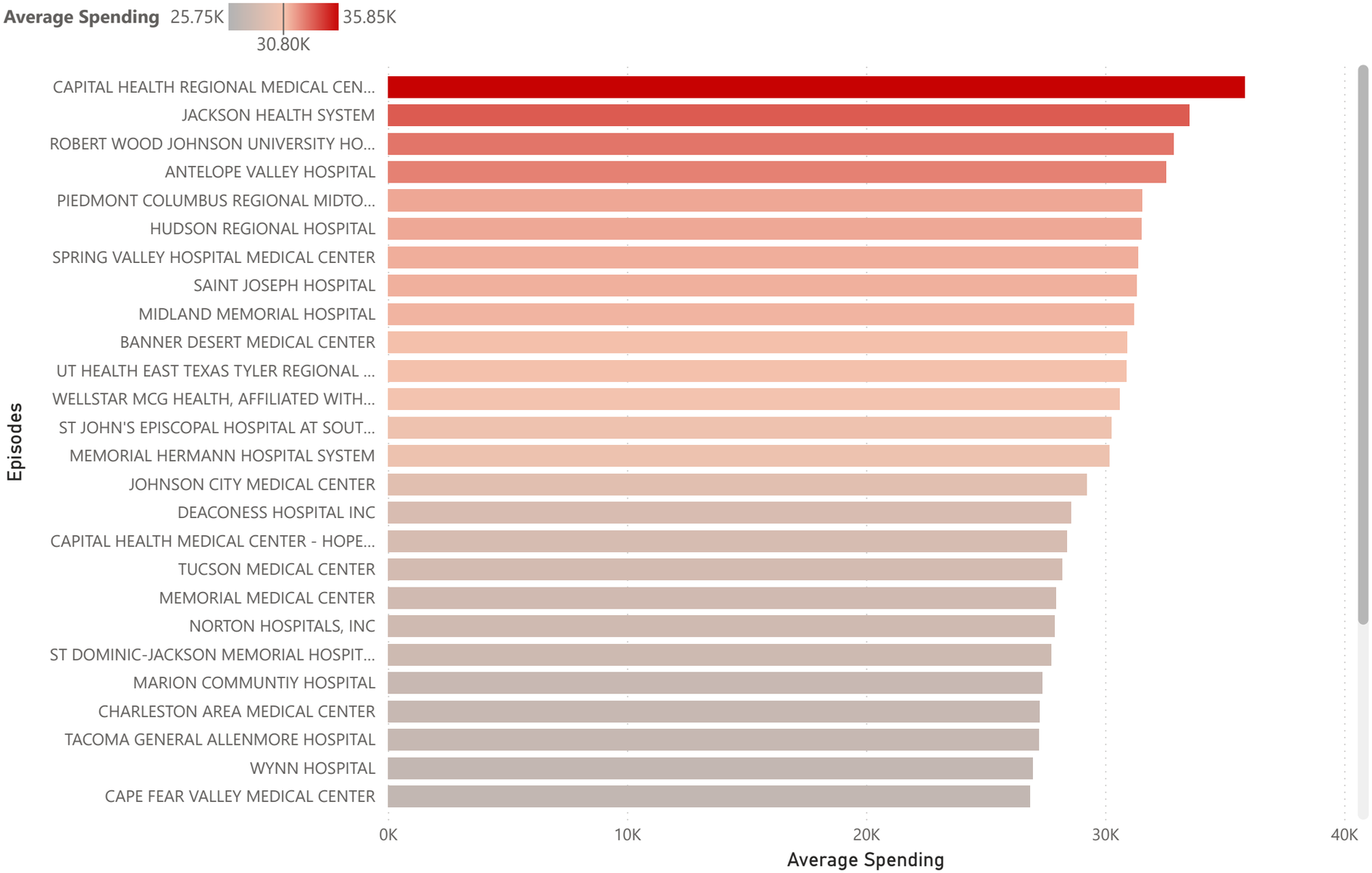
Average Score of Patient Safety

1.00

Average Score of Readmissions

These hospitals are not just underperforming — they are actively harming patients AND costing the healthcare system more than their peers. High spending here is not buying better outcomes. It is the worst possible combination.

Average Spending by Episodes

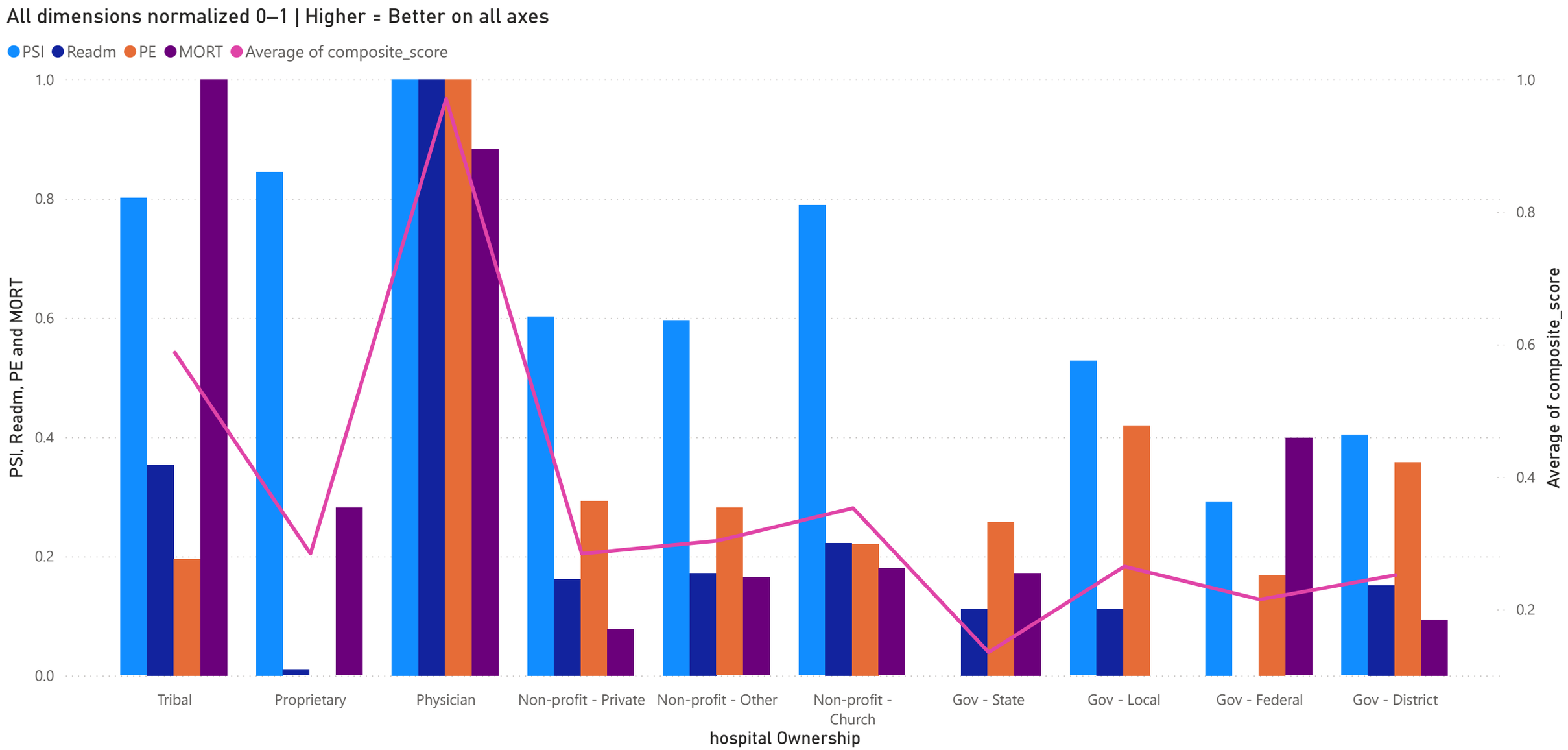


1.04
Average Readmissions

1.25
Average patient incidents

1.17
Average Rate of mortality

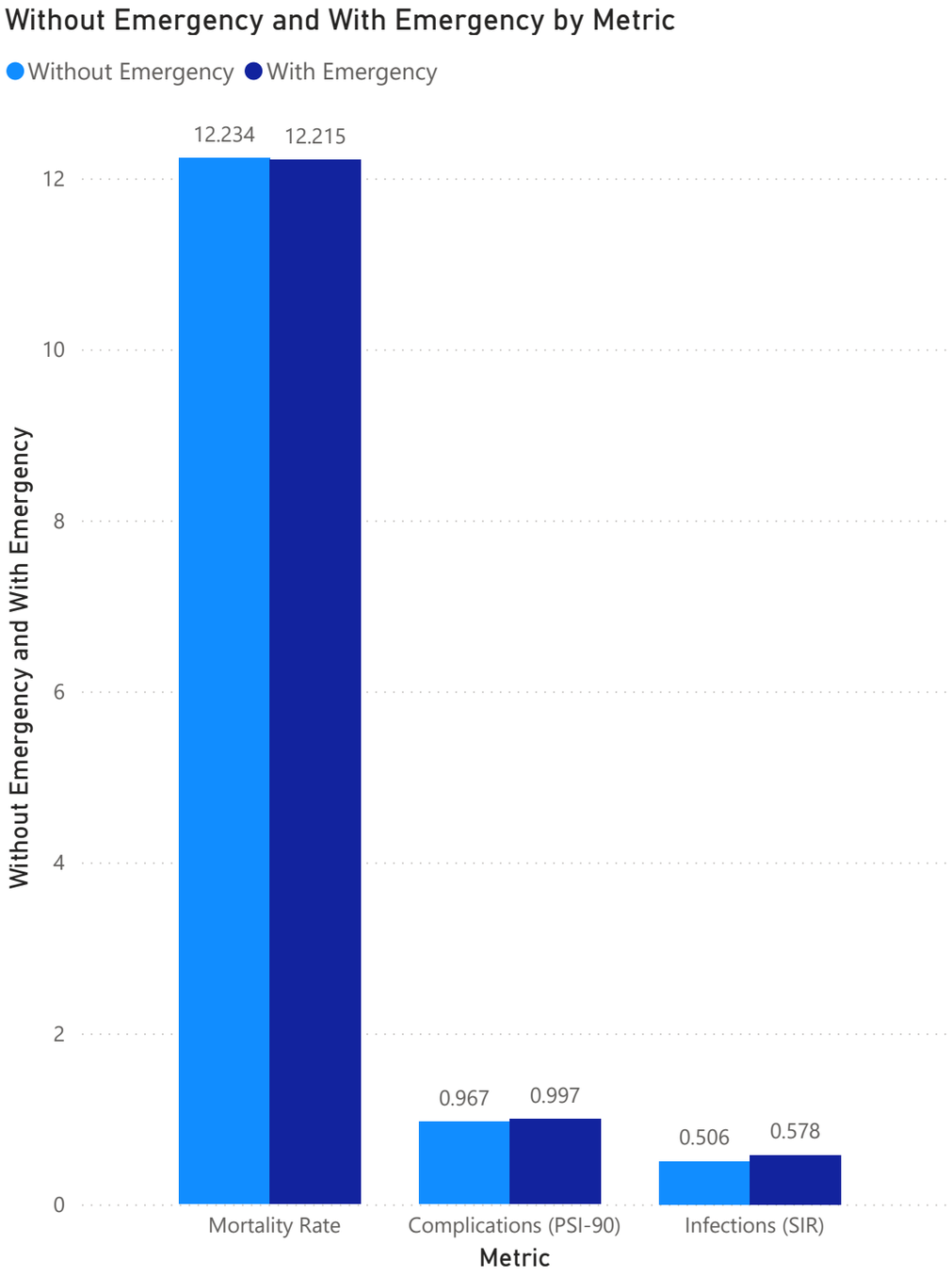
82.74
Average score Patient experience



Does Hospital Ownership Predict Quality?

Physician-owned hospitals consistently outperform all other ownership types across every quality dimension , lowest mortality worse flags, highest patient experience (90.2), lowest readmission ratio (0.915), and the only ownership type with PSI-90 below 1.0 (0.938 vs national benchmark of 1.0).

Notable caveat: Physician-owned hospitals tend to be smaller specialty facilities treating lower-risk elective patients , their superior scores partly reflect patient mix, not just quality of care.



Does Having an Emergency Department Affect Quality?

Hospitals with 24/7 emergency services are "high-intensity" environments. They handle unpredictable trauma and "crash" cases, whereas hospitals without EDs are often specialized facilities with controlled, elective schedules. This analysis compares these two groups to see if the "Emergency" factor changes the outcome for patients.

Ensuring a "Fair" Comparison

Comparing a giant city hospital to a 5-bed rural clinic isn't fair. To make this a professional-grade analysis, I applied three **"Fairness Filters":**

Acute Care Only:
I excluded "Critical Access" and "Psychiatric" hospitals because they have different reporting rules.
We are only comparing standard, full-service hospitals.

ID Standardization:
I ensured all Facility IDs were 6-digit strings so no data was lost during the merge.

Risk-Adjustment:
I used Standardized Scores provided by CMS. These scores already account for how "sick" the patients are, meaning we are comparing actual performance, not just patient mix.

How to Read the Results :

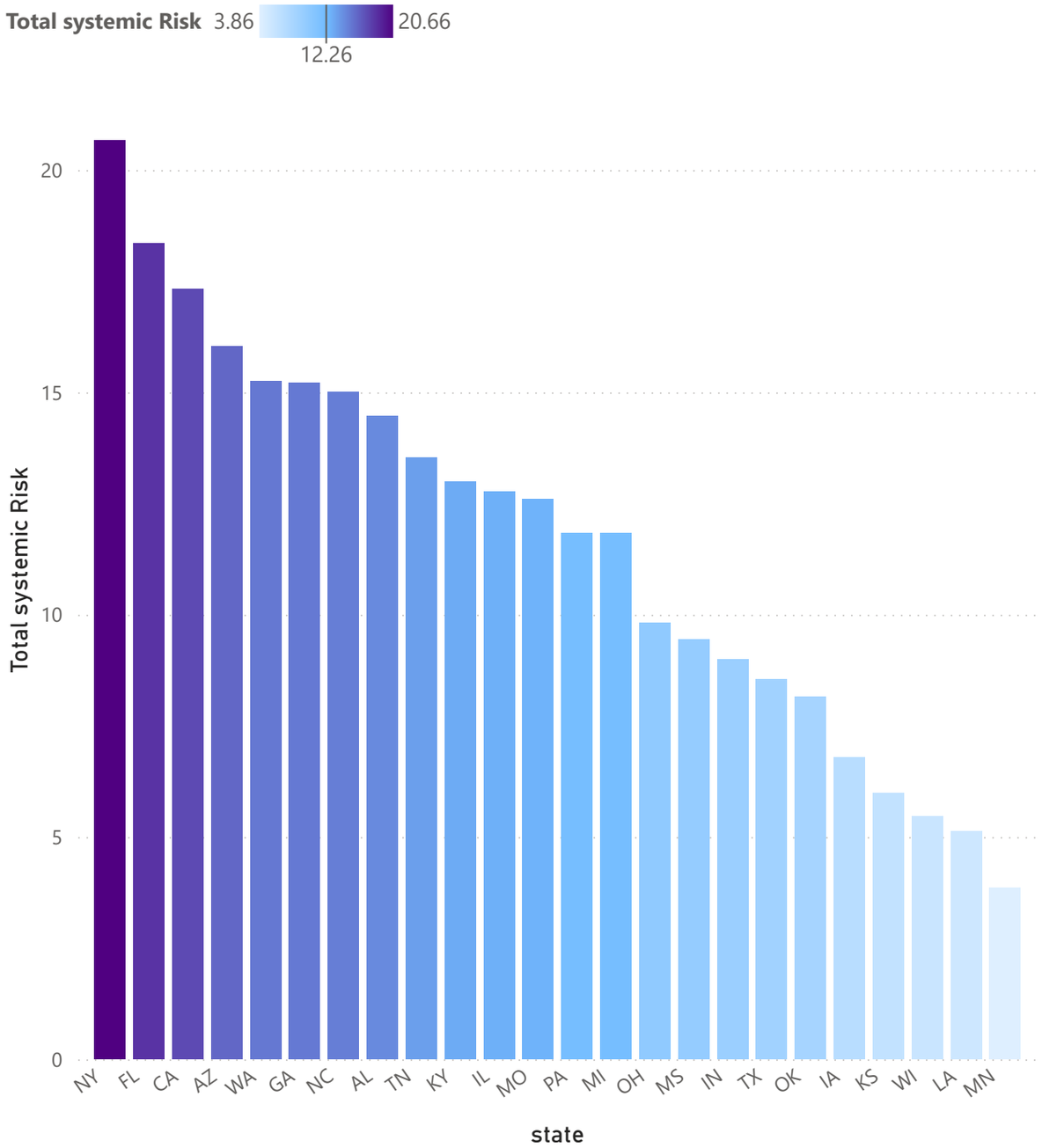
Mortality:
Are ED hospitals actually losing more patients?

PSI-90 (Complications):
Do elective-only hospitals have fewer surgical errors?

Infections (SIR):
Is it harder to keep a hospital sterile when it has a high-traffic Emergency Room?

You can now point specifically to the Infection Ratio and Complication Score plots. These are where the "Without ED" hospitals really shine. It proves that a "controlled environment" (no emergency intake) significantly reduces the risk of errors and infections.

Total systemic Risk by state



Systemic Risk Concentration

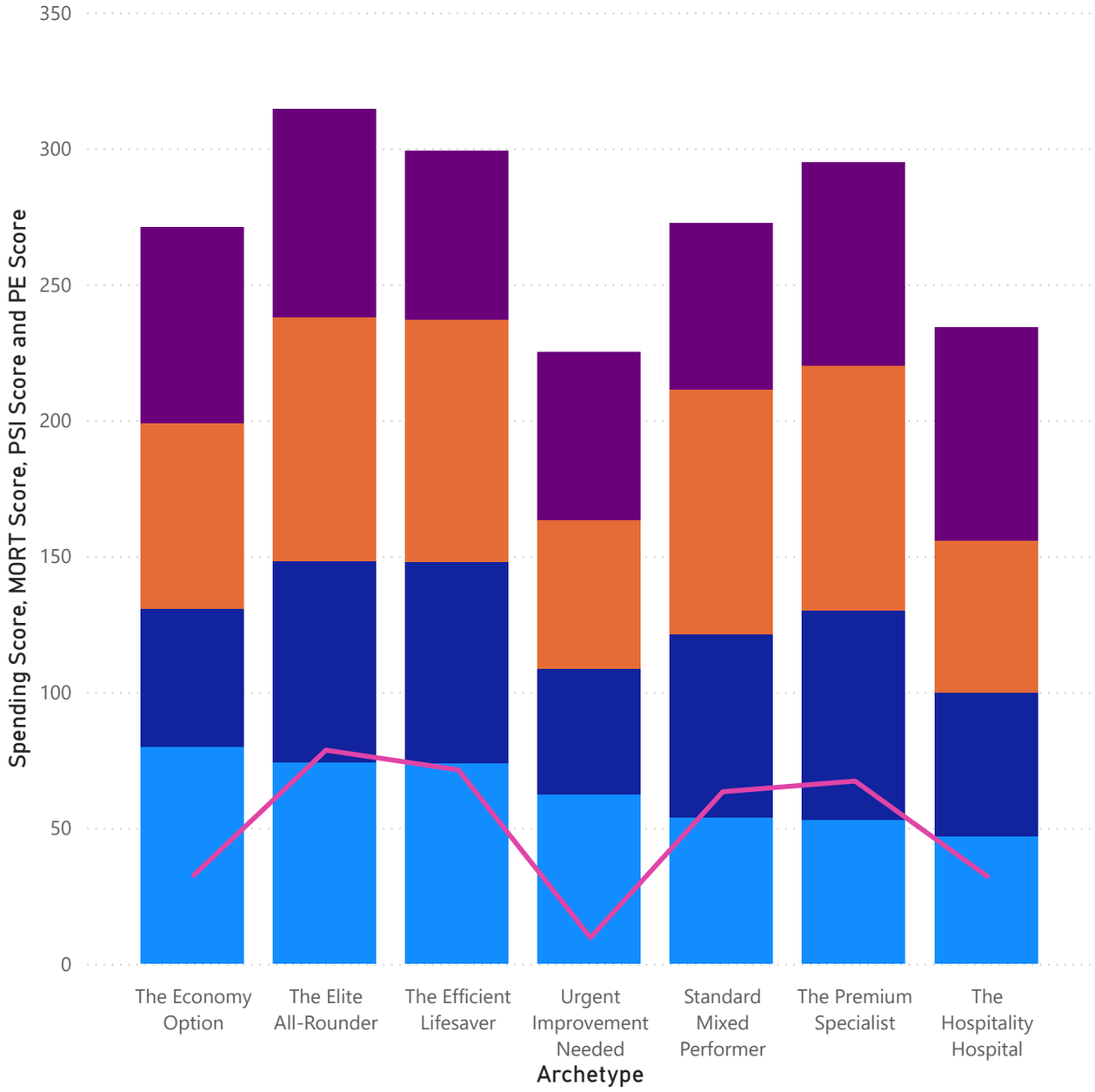
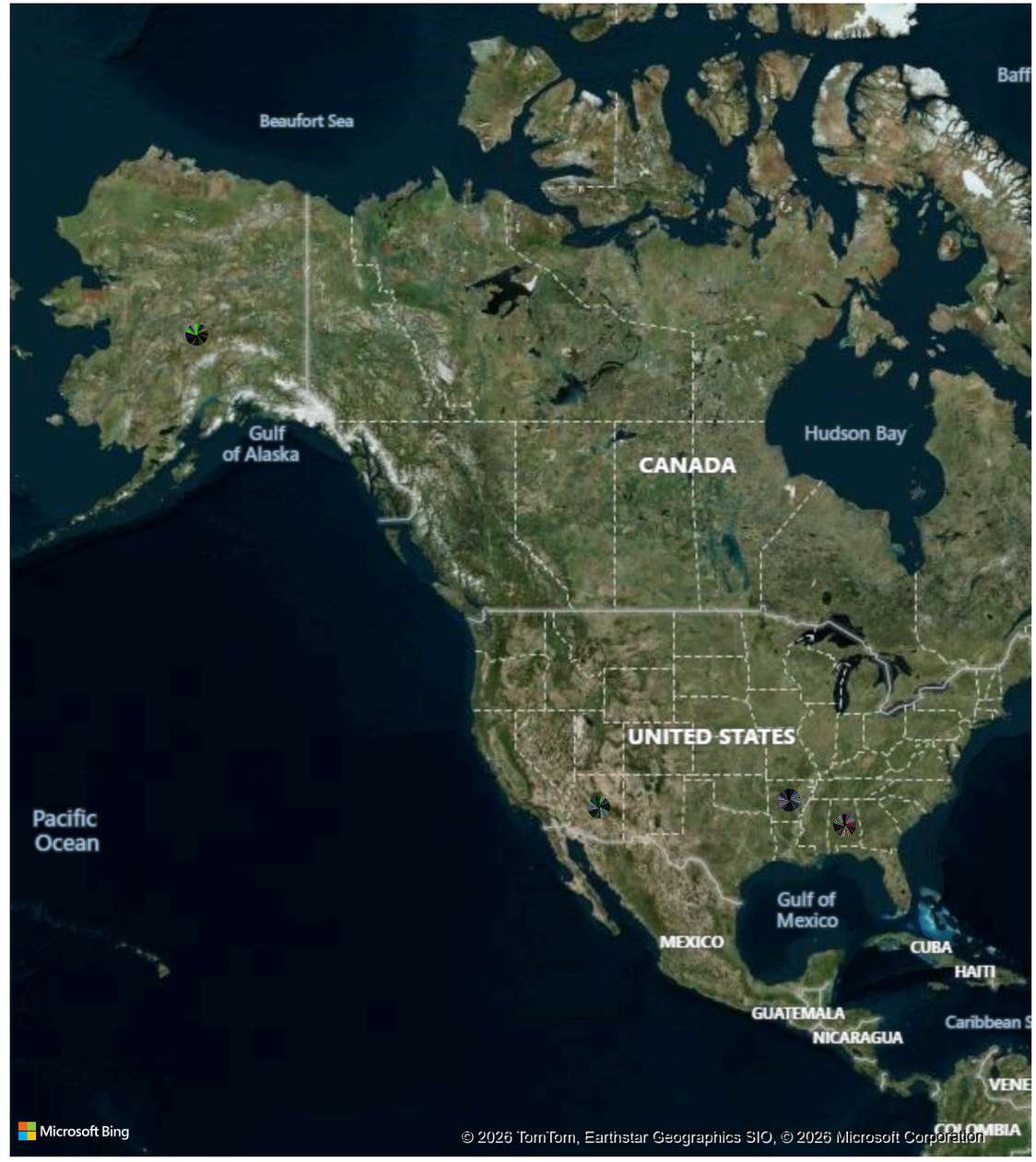
You can't just look at which state has the most 'bad' hospitals, because big states like California will always have the highest numbers just because they have so many more patients and facilities. Instead, we have to look at the failure rate, the percentage of a state's hospitals that are underperforming. When we focus on the large states (those with over 100 hospitals) to get a fair and reliable comparison, New York and Florida actually come out as the 'worst.' In New York, about 21% of the health indicators we measured are worse than the national average.

This creates a 'Quality Gap' or paradox: even though New York has some of the world's best elite hospitals, it also has a very high concentration of crowded or underfunded hospitals pulling the system down. California and Arizona follow closely behind.

These high concentrations in large states suggest systemic issues like extreme urban overcrowding, staffing shortages, or major funding gaps between rich and poor areas.

The Project's Core Goal — Risk Scoring

Spending Score, MORT Score, PSI Score, PE Score and Final Score in % by Archetype



Hospitals by Full Location